IT0614-MACHINE LEARNING **M. PRANATI**

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ASSESSMENT CO2 AT1

1. **A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 marks).**

Solution

Given:

- Study hours input, $\mathbf{x_1 = 2}$
- Attendance input, $\mathbf{x_2 = 1}$
- Weight for study hours, $\mathbf{w_1 = 0.6}$
- Weight for attendance, $\mathbf{w_2 = 0.4}$
- Bias, $\mathbf{b = -0.5}$

Step 1: Calculate the Net Input

The net input to the perceptron is:

Substitute the given values:

$$
\begin{aligned}
&= (0.6 \times 2) + (0.4 \times 1) + (-0.5) \\
&= 1.2 + 0.4 - 0.5 \\
&= 1.1
\end{aligned}
$$

Net Input = 1.1

Step 2: Apply the Step Activation Function

The step activation function is:

$$f(x) = \begin{cases} 1, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases}$$

Since the net input is:

$$1.1 \geq 0$$

The output is: 1

Step 3: Conclusion

- **Net Input = 1.1**
- **Perceptron Output = 1**
- **Decision:** The student **Passes**.

Final Answer (5 Marks)

- **Net Input:** (1.1)
- **Step Activation Output:** (1)
- **Conclusion:** Since the output is **1**, the perceptron classifies the student as **Pass**.

2. **In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)**
   Solution
   Given:

- Initial weight, **w = 0.5**
- Learning rate, **η = 0.1**
- Input, **x = 2**
- Actual output, **y = 0.6**
- Target output, **t = 1**

Step 1: Calculate the Error
The error is given by:

$$\begin{aligned} &\text{Error} = \text{Target Output} - \text{Actual Output} \\ &= 1 - 0.6 \\ &= 0.4 \end{aligned}$$

Error = 0.4

Step 2: Calculate the Weight Update (Delta Rule)
The delta rule is:

$$\Delta w = \eta \times (\text{Error}) \times x$$

Substitute the given values:

$$\begin{aligned} &\Delta w = 0.1 \times 0.4 \times 2 \\ &= 0.08 \end{aligned}$$

Weight Update (Δw) = 0.08

Step 3: Calculate the New Updated Weight

$$\text{New Weight} = \text{Old Weight} + \Delta w$$
$$= 0.5 + 0.08$$
$$= 0.58$$

New Updated Weight = 0.58

Conclusion

- Error = 0.4
- Weight Update = 0.08
- Updated Weight after one iteration = 0.58